Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering- Computer Science & Engineering **(Artificial Intelligence and Machine Learning)**		
Course Name	Programming Abstractions for AI	Session	July-Dec 2026
Course Code	24CAI0302	Semester/Batch	5^{th}/2024
L-T-P (Per Week)	2-0-4	Course Credits	4
Pre-requisite	Programming Fundamentals, Object-Oriented Programming	NHEQF Level	5.5
Course Coordinator	Ms. Rakhi	SDG Number	4,8,9

1. Objectives of the Course

The course Programming Abstractions for AI is designed to provide students with a strong foundation in programming concepts and data structures that are essential for developing intelligent software systems. The course emphasizes problem-solving, algorithmic thinking, and efficient implementation of computational solutions using Java. Students will learn to analyze the performance of algorithms, implement fundamental and advanced data structures, and apply programming abstractions to solve real-world and AI-related problems. By the end of the course, students will be equipped with the skills required to design optimized programs, manage data efficiently, and build a solid programming foundation for advanced Artificial Intelligence and Machine Learning courses.

The main objectives of the courses are to:

- To apply the concepts of data structure paradigm to analyse real life problems.
- To develop efficient solutions for logical problems using JAVA language.
- Exercise and reinforce prior programming knowledge to effectively code standard problem.
- To identify and remove bugs in a JAVA program.

2. Course Learning Outcomes (CLOs)

Course Learning Outcomes	CLOs	Program Outcomes	NHEQF Level Descriptor	No. of Lectures
CLO01	Implement the concept of object-oriented techniques and methodologies using Java.	PO1, PO2, PO3, PO5, PO11	Q1, Q2	**6**
CLO02	Applying Exception Handling and multithreading concepts for implementing a Robust Application in Java.	PO1, PO2, PO3, PO4, PO5, PO11	Q2, Q3	**16**
CLO03	Applying Exception Handling and multithreading concepts for implementing a Robust Application in Java.	PO2, PO3, PO4, PO5, PO8, PO11	Q2, Q3	**28**
CLO04	Implementing several Data structures using Collection Framework and using database connectivity for a complete java application.	PO1, PO2, PO3, PO5, PO9, PO10, PO11	Q3, Q4	**10**
Total Contact Hours				**60**

CLO-PO-PSO Mapping grid |Program outcomes (POs) and Program Specific Outcomes (PSOs) are available as a part of Academic Program Guide

CLOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CLO01	H			M	L		M				M	H	L	M
CLO02	M	M		H	M	H				M		M		L
CLO03	M		M	M	H		H				H	H	M	L
CLO04		L	H	H	H		M			M		M	H	H

*H=High, M=Medium, L=Low

3. Recommended Books (Reference Books/Textbooks)

B01:"Data Structures and Algorithms in Java" by Robert Lafore, 2nd Edition.
B02:"Data Structures and Algorithms Made Easy in Java" by Narasimha Karumanchi, 5th Edition.
B03: "Data Structures and Algorithms in Java" by Michael T. Goodrich and Roberto Tamassia, 6th Edition.
B04: "Java: The Complete Reference" by Herbert Schildt, 9th Edition.

4. Other readings & relevant websites

S.No.	Link of Journals, Magazines, websites and Research Papers
1	DSA Tutorial (w3schools.com)
2	"https://www.javatpoint.com/data-structures-in-java" "https://www.javatpoint.com/data-structures-in-java" "https://www.javatpoint.com/data-structures-in-java"Javatpoint
3	DSA using Java - Data Structures (tutorialspoint.com)
4	"https://archive.nptel.ac.in/courses/106/105/106105225/" Structure and algorithms using Java
5	Data Structures Tutorial - HYPERLINK "https://www.geeksforgeeks.org/data-structures/" HYPERLINK "https://www.geeksforgeeks.org/data-structures/" HYPERLINK "https://www.geeksforgeeks.org/data-structures/"GeeksforGeeks

5. Recommended Tools and Platforms

NetBeans, VS Code, Eclipse

6. Course Plan: Theory+ Lab

Theory Plan

Lect. No.	Topics
1-2	Data Structures and Algorithms: Basic Terminology, Elementary Data Organization, Data Structures and Operations
3-4	Algorithm: Complexity, Time and Space & Complexity, Asymptotic Notations for Complexity (Ω, ω, θ, O, o)
5-7	Array: Introduction, Representation of Linear Arrays in Memory, Traversing Linear Arrays, Insertion and Deletion in arrays.
8	Searching: Linear and Binary Search with their Complexity.
9-12	Character Arrays, Strings, Declaration, and Initialization of character array, Memory Representation, Basic and Advanced Operations. Sorting a Char Array
Continuous Evaluation-1 (Lectures 1-12)	

13-15	Bubble Sort Algorithm Selection Sort Algorithm, Insertion Sort Algorithm, Quick Sort, Merge Sort
16-18	Recursion & Backtracking: Introduction Recursion and Recursive Function
19-21	Deep Diving into recursion, Implementation of Recursion on Subsets, Backtracking Algorithms.
22-23	Time Complexity and Space Complexity, Analysis of Iterative and Recursive Algorithms
Continuous Evaluation-2 (Lectures 13- 23)	
24-26	Linked List: Introduction & its memory representation, traversing a Linked List, Insertion into Linked List (sorted and unsorted Linked List), Deleting from Linked List
27-29	Operations on Doubly Linked List, Circular linked List & its applications
30-31	Stacks: Array representation of Stacks, implementation of stack using linked list.
32-35	Applications: Arithmetic Expressions, Polish Notation, Transforming Infix Expressions into Postfix Expressions, Implementations of recursive and non-recursive procedures by Stacks
36-38	Queues: Representation as Array and Linked List.
39-40	Deques, Circular Queues, Priority Queues
41-44	Trees: Binary trees, complete binary trees, Binary Search Trees, Data structures for representing binary trees, Insertion, Deletion and Searching of Binary Search
45-48	Tree and their Implementation, Tree Traversal: preorder, In order, Post order and their algorithms, Their Implementation. B Trees.
Continuous Evaluation-3 (Lectures 24- 48)	
49-50	Heaps, Difference between heap and Array, insertion and deletion in heap, Heap sort implementation and its applications.
51	Graphs: Basic terminology, directed and undirected graphs, notion of path.
52-54	Representation of graphs: edge list structures, adjacency list structures, adjacency matrix, Linked List representation of Graph.
55-56	Operations on Graph, Graph traversals: DFS, BFS and their implementation.
57-58	Hashmap, Hashing Techniques, Collision and its resolving. Tree Data Structure, Basic Operations on Tree.
59	Introduction to Dynamic Programming, Standard problems on Dynamic Programming, One Dimensional Dynamic Programming
60	Two-Dimensional Dynamic Programming, Top DP Algorithms Greedy Algorithms
Continuous Evaluation-3 (49- 60 Lectures)	
END-TERM EXAM (FULL SYLLABUS)	

Lab Plan

Lab No.	Experiment (s)
1	Implement programs to demonstrate basic Java review, object-oriented concepts, classes, objects, methods, and packages.
2	Write programs to calculate time complexity of simple iterative algorithms and compare execution time for different input sizes.
3	Implement array operations including traversal, insertion, deletion, searching, and updating elements.
4	Implement Linear Search and Binary Search. Compare their execution time and complexity.
5	Write programs to perform character array and string operations including sorting, reversal, palindrome checking, and frequency count.

6	Implement Bubble Sort, Selection Sort, and Insertion Sort. Compare their performance on different datasets.
7	Implement Quick Sort and Merge Sort. Analyze divide-and-conquer strategy and compare complexities.
8	Write recursive programs for factorial, Fibonacci, GCD, Tower of Hanoi, and binary search.
9	Implement Backtracking algorithms such as N-Queens, Rat in a Maze, and Sudoku Solver.
10	Analyze recursive and iterative implementations by comparing time and space complexity.
11	Implement Singly Linked List operations: insertion, deletion, traversal, searching, and updating nodes.
12	Implement Doubly Linked List and Circular Linked List with insertion, deletion, and traversal operations.
13	Implement Stack using arrays and linked lists. Perform push, pop, peek, and display operations.
14	Implement applications of Stack including expression evaluation, infix-to-postfix conversion, postfix evaluation, and parentheses matching.
15	Implement Queue using arrays and linked lists. Perform enqueue, dequeue, front, rear, and display operations.
16	Implement Circular Queue, Deque, and Priority Queue with suitable applications.
17	Implement Binary Tree and Binary Search Tree. Perform insertion, deletion, searching, and tree traversals (Preorder, Inorder, Postorder).
18	Implement Heap operations including insertion, deletion, heap construction, and Heap Sort.
19	Implement Graph representations using adjacency matrix and adjacency list. Perform DFS and BFS traversals.
20	Implement HashMap operations, collision handling techniques, and Trie data structure for efficient searching.
21	Implement Dynamic Programming problems such as Fibonacci using memoization, 0/1 Knapsack, and Longest Common Subsequence.
22	Implement Greedy Algorithms such as Activity Selection, Fractional Knapsack, and Huffman Coding. Conduct mini-project demonstration integrating multiple data structures.

7. Delivery/Instructional Resources

Plan (Theory + Lab)

Lec. No.	Topics	CLO	Book & Chapter	TLM	ALM	Web References	Audio-Video
1-2	Data Structures and Algorithms: Basic Terminology, Elementary Data Organization, Data Structures and Operations	CLO01	Book1 Ch1	PPT	Think-Pair-Share	https://www.tutorialspoint.com/data_structures_algorithms/data_structure_overview.htm	https://www.youtube.com/results?search_query=data+structures+introduction
3-4	Algorithm: Complexity, Time and Space & Complexity, Asymptotic Notations for Complexity (Ω, ω, θ, O, o)	CLO01	Book1 Ch2	PPT	Problem Solving	https://www.geeksforgeeks.org/analysis-algorithms-big-o-analysis/	https://www.youtube.com/results?search_query=big+o+notation

5-7	Array: Introduction, Representation of Linear Arrays in Memory, Traversing Linear Arrays, Insertion and Deletion in arrays.	CLO01	Book1 Ch3	PPT + White Board	Hands-On Coding	https://www.geeksforgeeks.org/array-data-structure/	https://www.youtube.com/results?search_query=arrays+data+structure
8	Searching: Linear and Binary Search with their Complexity.	CLO02	Book1 Ch3	PPT	Live Coding	https://www.geeksforgeeks.org/searching-algorithms/	https://www.youtube.com/results?search_query=linear+binary+search
9-12	Character Arrays, Strings, Declaration, and Initialization of character array, Memory Representation, Basic and Advanced Operations. Sorting a Char Array	CLO02	Book4 Ch7	PPT	Coding Exercise	https://www.geeksforgeeks.org/java-string/	https://www.youtube.com/results?search_query=java+strings
13-15	Bubble Sort Algorithm Selection Sort Algorithm, Insertion Sort Algorithm, Quick Sort, Merge Sort	CLO02	Book1 Ch7	PPT	Comparative Analysis	https://www.geeksforgeeks.org/sorting-algorithms/	https://www.youtube.com/results?search_query=sorting+algorithms
16-18	Recursion & Backtracking: Introduction Recursion and Recursive Function	CLO02	Book1 Ch6	PPT	Think-Pair-Share	https://www.geeksforgeeks.org/recursion/	https://www.youtube.com/results?search_query=recursion
19-21	Deep Diving into recursion, Implementation of Recursion on Subsets, Backtracking Algorithms.	CLO02	Book2 Ch18	PPT	Coding Exercise	https://www.geeksforgeeks.org/backtracking-algorithms/	https://www.youtube.com/results?search_query=backtracking
22-23	Time Complexity and Space Complexity, Analysis of Iterative and	CLO02	Book1 Ch2	PPT	Problem Solving	https://www.geeksforgeeks.org/time-complexity-and-space-complexity/	https://www.youtube.com/results?search_query=time+spa

	Recursive Algorithms						ce+complexity
24-26	Linked List: Introduction & its memory representation, traversing a Linked List, Insertion into Linked List (sorted and unsorted Linked List), Deleting from Linked List	CLO03	Book1 Ch5	PPT	Coding Exercise	https://www.geeksforgeeks.org/data-structures/linked-list/	https://www.youtube.com/results?search_query=linked+list
27-29	Operations on Doubly Linked List, Circular linked List & its applications	CLO03	Book1 Ch5	PPT	Demonstration	https://www.geeksforgeeks.org/doubly-linked-list/	https://www.youtube.com/results?search_query=doubly+linked+list
30-31	Stacks: Array representation of Stacks, implementation of stack using linked list.	CLO03	Book1 Ch4	PPT	Coding Exercise	https://www.geeksforgeeks.org/stack-data-structure/	https://www.youtube.com/results?search_query=stack+data+structure
32-35	Applications: Arithmetic Expressions, Polish Notation, Transforming Infix Expressions into Postfix Expressions, Implementations of recursive and non-recursive procedures by Stacks	CLO03	Book1 Ch4	PPT	Problem Solving	https://www.geeksforgeeks.org/stack-applications/	https://www.youtube.com/results?search_query=infix+to+postfix
36-38	Queues: Representation as Array and Linked List.	CLO03	Book1 Ch5	PPT	Coding Exercise	https://www.geeksforgeeks.org/queue-data-structure/	https://www.youtube.com/results?search_query=queue+data+structure
39-40	Deques, Circular Queues, Priority Queues	CLO03	Book2 Ch8	PPT	Group Discussion	https://www.geeksforgeeks.org/deque-data-structure/	https://www.youtube.com/results?search_query=priority+queue

CHITKARA UNIVERSITY

41-44	Trees: Binary trees, complete binary trees, Binary Search Trees, Data structures for representing binary trees, Insertion, Deletion and Searching of Binary Search	CLO04	Book1 Ch8	PPT	Think-Pair-Share	https://www.geeksforgeeks.org/binary-search-tree-data-structure/	https://www.youtube.com/results?search_query=binary+search+tree
45-48	Tree and their Implementation, Tree Traversal: preorder, In order, Post order and their algorithms, Their Implementation. B Trees.	CLO04	Book1 Ch8	PPT	Coding Exercise	https://www.geeksforgeeks.org/tree-traversals-inorder-preorder-and-postorder/	https://www.youtube.com/results?search_query=tree+traversal
49-50	Heaps, Difference between heap and Array, insertion and deletion in heap, Heap sort implementation and its applications.	CLO04	Book1 Ch9	PPT	Coding Exercise	https://www.geeksforgeeks.org/heap-data-structure/	https://www.youtube.com/results?search_query=heap+sort
51	Graphs: Basic terminology, directed and undirected graphs, notion of path.	CLO04	Book1 Ch13	PPT	Group Discussion	https://www.geeksforgeeks.org/graph-data-structure-and-algorithms/	https://www.youtube.com/results?search_query=graph+data+structure
52-54	Representation of graphs: edge list structures, adjacency list structures, adjacency matrix, Linked List representation of Graph.	CLO04	Book1 Ch13	PPT	Problem Solving	https://www.geeksforgeeks.org/graph-and-its-representations/	https://www.youtube.com/results?search_query=graph+representation
55-56	Operations on Graph, Graph traversals: DFS, BFS and their implementation.	CLO04	Book1 Ch13	PPT	Coding Exercise	https://www.geeksforgeeks.org/breadth-first-search-or-bfs-for-a-graph/	https://www.youtube.com/results?search_query=dfs+bfs
57-58	Hashmap, Hashing Techniques, Collision and its resolving. Trie Data Structure,	CLO04	Book2 Ch10	PPT	Case Study	https://www.geeksforgeeks.org/hashing-data-structure/	https://www.youtube.com/results?search_query=hashing+

	Basic Operations on Trie.						data+structure
59	Introduction to Dynamic Programming, Standard problems on Dynamic Programming, One-Dimensional Dynamic Programming	CLO04	Book2 Ch20	PPT	Coding Exercise	https://www.geeksforgeeks.org/dynamic-programming/	https://www.youtube.com/results?search_query=dynamic+programming
60	Two-Dimensional Dynamic Programming, Top DP Algorithms Greedy Algorithms	CLO04	Book2 Ch20	PPT	Problem Solving	https://www.geeksforgeeks.org/greedy-algorithms/	https://www.youtube.com/results?search_query=greedy+algorithm

8. Remedial Classes

After every Sessional Test, different types of learners will be identified, and special discussions will be planned and scheduled accordingly for the slow learners.

Learner Type-I	Learner Type- II	Learner Type- III
Remedial Classes, Doubt Sessions, Guided Tutorials	Workshop, Doubt Session	Project

9. Self-Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCS
1	Implement Maze Solver using Backtracking	CLO02, CLO03	Think–Pair–Share	https://www.geeksforgeeks.org/backtracking-algorithms/
2	Solve Pathfinding Problems using Graph Traversal (BFS & DFS)	CLO03, CLO04	Think–Pair–Share	https://www.geeksforgeeks.org/breadth-first-search-or-bfs-for-a-graph/
3	Implement LRU Cache using HashMap and Linked List	CLO04	Think–Pair–Share	https://www.geeksforgeeks.org/lru-cache-implementation/

10. Delivery Details of Content Beyond Syllabus

Any content delivered beyond the syllabus will be planned for all students, and the schedule will be notified accordingly.

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Introduction to Advanced Data Structures for AI (Trie, Disjoint Set Union, Segment Tree)	CLO04	PO2, PO3, PO11	Guided Coding Session	https://leetcode.com/explore/learn/
2	Introduction to Competitive Programming and Problem Solving Platforms (LeetCode, Codeforces)	CLO02, CLO03	PO1, PO2, PO4	Case Study & Brainstorming	https://www.geeksforgeeks.org/a-search-algorithm/

11. Evaluation Scheme & Components

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component 2	Sessional Tests (STs)	02*	40%	40	Computer Based Test	CLO01-CLO04
Summative	Component 3	End Term Examination	01**	60%	60	Computer Based Test	CLO01-CLO04
Total			**100%**				

**Best of the 02 STs will be considered for the evaluation of the STs as final marks.*
***To be eligible to appear for the End Term Exam, attendance must be at least 75%.*
**** The passing criterion for exam is 50% score of total score.*

12. Syllabus of the Course

Subject: Programming Abstractions for AI		**Subject Code:** 24CAI0302	
S. No.	**Topic (s)**	**No. of Lectures**	**Weightage %**
1	Data Structures and Algorithms: Basic Terminology, Elementary Data Organization, Data Structures and Operations	2	**3%**
2	Algorithm: Complexity, Time and Space & Complexity, Asymptotic Notations for Complexity (Ω, ω, θ, O, o)	2	**3%**
3	Array: Introduction, Representation of Linear Arrays in Memory, Traversing Linear Arrays, Insertion and Deletion in arrays. Searching: Linear and Binary Search with their Complexity. Character Arrays, Strings, Declaration, and Initialization of character array, Memory Representation, Basic and Advanced Operations. Sorting a Char Array	8	**14%**
4	Bubble Sort Algorithm Selection Sort Algorithm, Insertion Sort Algorithm, Quick Sort, Merge Sort	3	**5%**
5	Recursion & Backtracking: Introduction Recursion and Recursive Function	3	**5%**
6	Deep Diving into recursion, Implementation of Recursion on Subsets, Backtracking Algorithms.	3	**5%**
7	Time Complexity and Space Complexity, Analysis of Iterative and Recursive Algorithms	2	**3%**

8	Linked List: Introduction & its memory representation, traversing a Linked List, Insertion into Linked List (sorted and unsorted Linked List), Deleting from Linked List, Operations on Doubly Linked List, Circular linked List & its applications	6	**10%**
9	Stacks: Array representation of Stacks, implementation of stack using linked list.	2	**3%**
10	Applications: Arithmetic Expressions, Polish Notation, Transforming Infix Expressions into Postfix Expressions, implementations of recursive and non-recursive procedures by Stacks	4	**7%**
11	Queues: Representation as Array and Linked List, Deques, Circular Queues, Priority Queues, Deques	5	**8%**
12	Trees: Binary trees, complete binary trees, Binary Search Trees, Data structures for representing binary trees, Insertion, Deletion and Searching of Binary Search	4	**7%**
13	Tree and their Implementation, Tree Traversal: preorder, In order, Post order and their algorithms, Their Implementation of B Trees.	4	**7%**
14	Heaps, Difference between heap and Array, insertion and deletion in heap, Heap sort implementation and its applications.	2	**3%**
15	Graphs: Basic terminology, directed and undirected graphs, notion of path. Representation of graphs: edge list structures, adjacency list structures, adjacency matrix, Linked List, representation of Graph. Operations on Graph, Graph traversals: DFS, BFS and their implementation.	6	**10%**
16	Hashmap, Hashing Techniques, Collision and its resolving. Tree Data Structure, Basic Operations on Trie.	2	**3%**
17	Introduction to Dynamic Programming, Standard problems on Dynamic Programming, One-Dimensional Dynamic Programming	1	**2%**
18	Two-Dimensional Dynamic Programming, Top DP Algorithms Greedy Algorithms	1	**2%**

13. Academic Integrity Policy

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by

Designation	**Name**	**Signature**
Course Coordinator	Ms. Rakhi	
Head Academic Delivery	Dr. Vandana Sood	
Dean (CSE - AI)	Dr. Sushil Kumar Narang	
Date	23/06/2026	